

Frédéric Andrian

Software Engineer · Full-Stack Development and AI

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Profile

Recent engineering graduate from ENSC, Bordeaux INP, available as soon as possible. At INRIA (French National Institute for Research in Digital Science and Technology), I independently built a working AI application from user needs to implementation, combining a Python backend, a React interface and a multi-step agent. Academic and personal projects have broadened my software development experience. I want to build useful AI-powered products at Bending Spoons, take ownership of technical challenges and continue learning in an international team.

Experience

INRIA

Bordeaux, France

AI Research Engineering Intern

Mar. 2026 – Sept. 2026

- Independently designed and built a working application and AI agent to help researchers find scientific papers, with guidance from my supervisors.
- Developed a Python/FastAPI backend with REST APIs documented using OpenAPI and MongoDB persistence.
- Built a React/Next.js interface for interacting with the agent and its tools, including user selection of messages to include in the model context.
- Implemented a multi-step agent without an orchestration framework to understand its underlying mechanisms, integrating the OpenAI API, prompts, tool execution, state and memory management.
- Implemented a RAG pipeline with document chunking, embeddings and semantic retrieval using ChromaDB.
- Automated online scientific-paper metadata collection in Python and delivered cron task results through webhooks.
- Containerized components with Docker and set up a GitHub CI/CD pipeline. Managed a web application using LLMs installed locally on a Mac Studio, including model installation and configuration.
- Used vLLM to serve models on HPC infrastructure with NVIDIA GPUs. Compared Qwen, Gemma and GLM and quantization options using Slurm, measuring latency, TTFT, response quality and context length.
- Defined the experimental protocol, analyzed results and worked on writing a scientific article comparing language models.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

June 2025 – Sept. 2025

- Developed a reinforcement learning model and its training environment for autonomous navigation of a hospital robot.
- Deployed and optimized the software and model on a Raspberry Pi for real-world testing.

Academic Projects

University Management REST API

ENSC, Bordeaux INP

C#, ASP.NET Core, Entity Framework Core, SQLite

- Developed a REST API in C# and ASP.NET Core for university data management.
- Used DTOs to structure the data exchanged through the API.
- Implemented data persistence with Entity Framework Core and a SQLite SQL database.
- Documented the API using Swagger/OpenAPI.

League of Legends Player Assistant

Academic project

Python, Pandas, scikit-learn, Random Forest, Discord

- Collected match data through an API to build a dataset for a player decision-support model.
- Cleaned and preprocessed the collected data using Pandas for model training.
- Trained a Random Forest model with scikit-learn to help players make decisions during a match.
- Integrated model predictions into a Discord bot to make the results accessible to players.

Skills

Development APIs and databases Agents and LLMs

Python, JavaScript, Java, C#, .NET, FastAPI, React, Next.js, Node.js
REST, OpenAPI, webhooks, SQL, SQLite, MongoDB, Entity Framework Core
OpenAI API, LangChain, vLLM, prompts, tool calling, multi-step agents, memory, context, RAG

ML and LLM evaluation Engineering tools Languages

PyTorch, TensorFlow, scikit-learn, Pandas, quantization, latency, response quality
Git/GitHub, Docker, CI/CD, Linux, cron tasks, HPC, Slurm
French (native), English (fluent, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, AI and human-computer interaction, with academic projects in C#/.NET.